

Test 1 Review Practice

MATH 130 · Wednesday, September 23 · Test 1: Friday, September 25

Name _____

No formulas are provided on Test 1. Work these problems without notes, then complete the formula page from memory.

Show every step. Give exact answers (fractions and radicals, not decimals). Check solutions in the original equation.

A · Algebra foundations

A1 Factor completely.

$$5x^2 + 5x - 30$$

A2 Factor completely.

$$3x^4 - 21x^3 + 30x^2$$

A3 Simplify. Use positive exponents only; all variables are nonzero.

$$\frac{x^{-4}y^6z^{-2}}{x^3y^{-2}z^5}$$

A4 Simplify. Use positive exponents only; all variables are nonzero.

$$\left(\frac{9xy^{-2}}{3z^3}\right)^3$$

A5 Subtract and simplify.

$$\frac{5}{2x} - \frac{3}{x+4}$$

A6 Subtract and simplify.

$$\frac{4}{5b} - \frac{3}{2}$$

B · Sets Write interval answers in interval notation.

B1–B3 Let

$$A: -4 \leq x < 3$$

$$B: x > 1$$

$$C: x \leq -2$$

$$B1. A \cap B \quad B2. B \cup C \quad B3. A \cup C$$

B4–B7 Let

$$P = \{2, 3, 5, 7, 11\}$$

$$Q = \{1, 2, 3, 4, 5\}$$

$$R = \{3, 6, 9, 12\}$$

$$S = \{4, 6, 8\}$$

$$B4. P \cup Q \quad B5. P \cap Q \quad B6. P \cap R \quad B7. P \cap S$$

C · Equations Solve for x. State any restrictions. Check solutions in the original equation.

C1

$$\frac{2}{x-1} + 1 = \frac{4}{x^2-1}$$

C2

$$6x^2 = 7x + 3$$

C3

$$\frac{1}{5}x^2 + x = \frac{12}{5}x - 2$$

C4

$$\frac{10}{x+3} = \frac{6}{x-1}$$

D · Coordinate geometry Give exact answers.

D1–D3 For the points $P(-2, 5)$ and $Q(6, -1)$, find

D1. the midpoint of PQ D2. the distance PQ D3. the equation of line PQ

D4 Write an equation of the circle with center $(-3, 1)$ and radius 7.

D5–D6 For the line $y = -2x + 7$ and the point $(4, -3)$, write a line through the point that is

D5. parallel D6. perpendicular

E · Maximum height Section 3.1

A ball is thrown straight up. Its height h , in feet, after t seconds is given. Find the maximum height.

E1

$$h(t) = 64t - 16t^2$$

E2

$$h(t) = -16t^2 + 96t + 6$$

Formulas from memory No formulas are provided on the test. Fill in each row without notes, then check.

Midpoint	
Distance	
Slope	
Point-slope form	
Parallel and perpendicular slopes	
Circle (center and radius)	
Time of maximum height	
Exponent rules (quotient, negative, power)	
Quadratic formula	

Answers Check after you finish. Worked solutions are on the lecture slides.

A · Algebra foundations

A1 $5(x + 3)(x - 2)$

A2 $3x^2(x - 2)(x - 5)$

A3 $\frac{y^8}{x^7z^7}$

A4 $\frac{27x^3}{y^6z^9}$

A5 $\frac{20 - x}{2x(x + 4)}$

A6 $\frac{8 - 15b}{10b}$

B · Sets

B1 $(1, 3)$

B2 $(-\infty, -2] \cup (1, \infty)$

B3 $(-\infty, 3)$

B4 $\{1, 2, 3, 4, 5, 7, 11\}$

B5 $\{2, 3, 5\}$

B6 $\{3\}$

B7 $\emptyset$

C · Equations

C1 $x = -3$ ($x \neq \pm 1$; $x = 1$ is excluded)

C2 $x = -1/3$ or $x = 3/2$

C3 $x = 2$ or $x = 5$

C4 $x = 7$ ($x \neq -3$, $x \neq 1$)

D · Coordinate geometry

D1 $(2, 2)$

D2 10

D3 $y = -(3/4)x + 7/2$

D4 $(x + 3)^2 + (y - 1)^2 = 49$

D5 $y = -2x + 5$

D6 $y = (1/2)x - 5$

E · Maximum height

E1 64 ft (at $t = 2$ s)

E2 150 ft (at $t = 3$ s)